

Mohamed Riyasudeen D

Embedded Software Engineer

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SUMMARY

Embedded Software Engineer with over 3 years of experience specializing in automotive Battery Management System (BMS) firmware development. Expertise includes NXP S32K322 microcontrollers, functional safety standards (ISO 26262), diagnostics, and hardware debugging. Skilled in embedded C programming, real-time operating systems (RTOS), and automotive communication protocols, including CAN. Proven ability to collaborate effectively with cross-functional teams to deliver robust, safety-critical embedded solutions. Dedicated to driving innovation in the electric vehicle (EV) industry by leveraging technical proficiency to optimize battery management systems and enhance overall system reliability.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer

iElektron Technologies (Onsite at Royal Enfield)

10/2023 – Present
Chennai

- Developed production CDD drivers for ADBMS6832 (cell monitor) & ADBM2950 (pack monitor) with functional safety; implemented cell voltage, NTC temperature, shunt current, open-wire detection, and IMD isolation calculations.
- Implemented passive cell balancing logic (voltage thresholds, PWMcontrol) in charge and sleep modes; reduced cell delta voltage by 10-20 mV during charge cycles & below 10 mV in sleep mode.
- Developed driver software for FS26-based power management and enabled Low Power Cell Monitoring to detect abnormal battery conditions in low-power mode and wake the controller when required.
- Worked on power management in BMS using PMIC (FS26) and used LPCM of AFEs to monitor cell voltages & temperature in low power mode.
- Developed software for external EEPROM, external flash, contactor control through high-side drivers, ORing protection IC control, and MCU peripherals such as ADC, timers, interrupts, GPIOs, and FreeRTOS.
- Worked on functional safety-oriented diagnostics, including open-wire detection, isolation-related monitoring, short-circuit and fault-oriented handling in BMS software.
- Used CANoe, CANalyzer, CAPL scripts, CAPL DLLs, Freemaster, logic analyzer, and log analysis tools during integration, debugging, validation, and fault analysis activities
- Built Vector CANoe panels + CAPL automation for BMS parameter monitoring (single-view dashboard) and fault injection; accelerated validation by 40%.
- Supported AIS-156 homologation and EMI/EMC/ESD testing for EV BMS at ARAI Pune and TUV Bangalore, with hands-on understanding of test cases, procedures, and terminology.

Embedded Software Engineer

iElektron Technologies

02/2023 – 09/2023
Chennai

- Developed a complete NVM manager for EEPROM emulation, enhancing memory management for data flash and code flash on Renesas RL78/F13.
- Implemented UDS services including Security Access, Read Data by Identifier, and others for vehicle diagnostics.
- Collaborated on bootloader development ensuring robust diagnostics for automotive applications.

EDUCATION

B.Tech /B.E - Electrical & Electronics Engineer

Syed Ammal Engineering College

08/2016 – 04/2020
Ramanathapuram

COURSES

Embedded Systems

Vector India

07/2022 – 02/2023
Chennai

SKILLS

MCU Platforms

S32K3x (NXP), Renesas RL78/F13, STM32

Programming

C/Embedded C, FreeRTOS, CAPL scripting, CAPL DLL, C++, Python (automation/scripts)

Comms/Protocols

CAN, SPI, I2C, UART, UDS (ISO 14229)

Debugging/Testing

Oscilloscope, Function generator, Regulated power supplies (RPS), Electronic loads, bidirectional supplies, board bring-up, Schematic analysis/review

Development Tools

S32 Design Studio, NXP FreeMASTER, Renesas e2 Studio, IAR IDE, Vector CANoe, Git (GitHub, Bitbucket), Qt Creator, VS Code, Visual Studio (DLL development)

Quality

MISRA C, Unit Testing, Functional Safety, AIS-156 (EV), EMI/EMC

Peripherals

ADC, Timers, GPIO, RTC, Interrupt Controllers, Watchdog

PROJECTS

EV BMS Firmware

Royal Enfield (NXP S32K322)

Battery pack data acquisition & calculation, Passive cell balancing, LPCM, FS26 PMIC power management, diagnostics, CANoe validation.

10/2023 – Present

Bootloader & UDS Diagnostics

Spark Minda (Renesas RL78/F13)

EEPROM emulation, NVM manager, UDS services (0x10, 0x27, 0x22, 0x2E, 0x34/36/37).

03/2022 – 09/2022

BINGO GAME

Console-based number game (Vector India Institute project)

- Implemented a console-based Bingo game in C during the Embedded Systems course at Vector India Institute.
- Designed the game logic for number generation, board updates, win detection, and score handling using structured C and modular functions.
- Used arrays and state machines to track called numbers and validate row/column/diagonal “Bingo” conditions.
- Focused on clean code, input validation, and separation of UI and core game logic to match embedded coding practices.

09/2022 – 10/2022